number of samples, we find that this entire procedure takes roughly 150 clock cycles per pixel. For a 40 MHz ADSP-2106, this corresponds to a data throughput of about 260k samples/second.

Comparing these different techniques in Fig. 28-11, we can make an important observation. *Nearly all DSP techniques require between 4 and 400 instructions (clock cycles in the SHARC family) to execute*. For a SHARC DSP operating at 40 MHz, we can immediately conclude that its data throughput will be between 100k and 10M samples per second, depending on how complex of algorithm is used.

Now that we understand how fast DSPs *can* process digitized signals, let's turn our attention to the other end; how fast do we *need* to process the data? Of course, this depends on the application. We will look at two of the most common, audio and video processing.

The data rate needed for an audio signal depends on the required quality of the reproduced sound. At the low end, telephone quality speech only requires capturing the frequencies between about 100 Hz and 3.2 kHz, dictating a sampling rate of about 8k samples/second. In comparison, high fidelity music must contain the full 20 Hz to 20 kHz range of human hearing. A 44.1 kHz sampling rate is often used for both the left and right channels, making the complete Hi Fi signal 88.2k samples/second. How does the SHARC family compare with these requirements? As shown in Fig. 28-11, it can easily handle high fidelity audio, or process several dozen voice signals at the same time.

Video signals are a different story; they require about *one-thousand* times the data rate of audio signals. A good example of low quality video is the the CIF (Common Interface Format) standard for videophones. This uses 352×288 pixels, with 3 colors per pixel, and 30 frames per second, for a total data rate of 9.1 million samples per second. At the high end of quality there is HDTV (high-definition television), using 1920×1080 pixels, with 3 colors per pixel, and 30 frames per second. This requires a data rate to over 186 million samples per second. These data rates are above the capabilities of a single SHARC DSP, as shown in Fig. 28-11. There are other applications that also require these very high data rates, for instance, radar, sonar, and military uses such as missile guidance.

To handle these high-power tasks, several DSPs can be combined into a single system. This is called **multiprocessing** or **parallel processing**. The SHARC DSPs were designed with this type of multiprocessing in mind, and include special features to make it as easy as possible. For instance, no external hardware logic is required to connect the external busses of multiple SHARC DSPs together; all of the bus arbitration logic is already contained within each device. As an alternative, the link ports (4 bit, parallel) can be used to connect multiple processors in various configurations. Figure 28-12 shows typical ways that the SHARC DSPs can be arranged in multiprocessing systems. In Fig. (a), the algorithm is broken into sequential steps, with each processor performing one of the steps in an "assembly line"